Highly accurate protein structure prediction with AlphaFold

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Proteins are essential to life, and understanding their structure can facilitate a mechanistic understanding of their function. Through an enormous experimental effort[1–4], the structures of around 100,000 unique proteins have been determined[5], but this represents a small fraction of the billions of known protein sequences[6,7]. Structural coverage is bottlenecked by the months to years of painstaking effort required to determine a single protein structure. Accurate computational approaches are needed to address this gap and to enable large-scale structural bioinformatics. Predicting the three-dimensional structure that a protein will adopt based solely on its amino acid sequence—the structure prediction component of the 'protein folding problem'[8]—has been an important open research problem for more than 50 years[9]. Despite recent progress[10–14], existing methods fall far short of atomic accuracy, especially when no homologous structure is available. Here we provide the first computational method that can regularly predict protein structures with atomic accuracy even in cases in which no similar structure is known. We validated an entirely redesigned version of our neural network-based model, AlphaFold, in the challenging 14th Critical Assessment of protein Structure Prediction (CASP14)[15], demonstrating accuracy competitive with experimental structures in a majority of cases and greatly outperforming other methods. Underpinning the latest version of AlphaFold is a novel machine learning approach that incorporates physical and biological knowledge about protein structure, leveraging multi-sequence alignments, into the design of the deep learning algorithm.

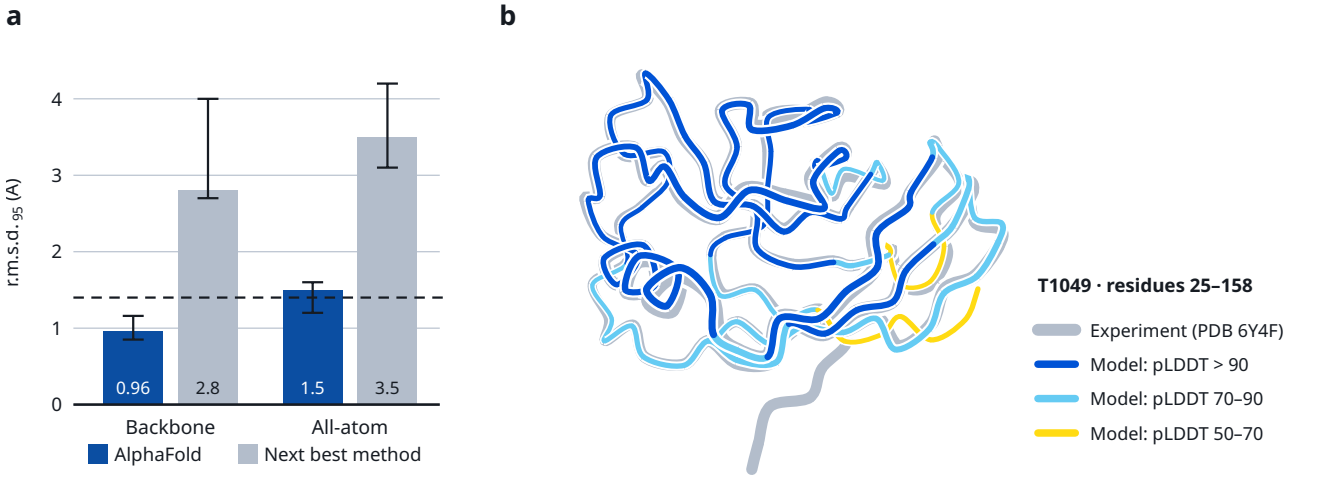


Fig. 1 | AlphaFold produces highly accurate structures. a, Median accuracy on the CASP14 domains of AlphaFold and of the next best method, as r.m.s.d.$_{95}$ over Cα atoms (backbone) and over all atoms, with 95% confidence intervals; the dashed line is the width of a carbon atom, about 1.4 Å. **b**, CASP14 target T1049 (PDB 6Y4F): the experimental Cα trace (grey) and the AlphaFold model of the same chain, coloured by its per-residue confidence (pLDDT).

Redrawn for this edition. a, from the values given in the text (the original plots the top 15 of 146 entries). b, from PDB 6Y4F (CC0) and AlphaFold DB model AF-B4EUK6-F1, AlphaFold Monomer v2.0 (CC BY 4.0), superposed on 134 Cα atoms; the original shows the CASP14 prediction.

The development of computational methods to predict three-dimensional (3D) protein structures from the protein sequence has proceeded along two complementary paths that focus on either the physical interactions or the evolutionary history. The physical interaction programme heavily integrates our understanding of molecular driving forces into either thermodynamic or kinetic simulation of protein physics[16] or statistical approximations thereof[17]. Although theoretically very appealing, this approach has proved highly challenging

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for even moderate-sized proteins due to the computational intractability of molecular simulation, the context dependence of protein stability and the difficulty of producing sufficiently accurate models of protein physics. The evolutionary programme has provided an alternative in recent years, in which the constraints on protein structure are derived from bioinformatics analysis of the evolutionary history of proteins, homology to solved structures[18,19] and pairwise evolutionary correlations[20–24]. This bioinformatics approach has benefited greatly from the steady growth of experimental protein structures deposited in the Protein Data Bank (PDB)[5], the explosion of genomic sequencing and the rapid development of deep learning techniques to interpret these correlations. Despite these advances, contemporary physical and evolutionary-history-based approaches produce predictions that are far short of experimental accuracy in the majority of cases in which a close homologue has not been solved experimentally and this has limited their utility for many biological applications.

In this study, we develop the first, to our knowledge, computational approach capable of predicting protein structures to near experimental accuracy in a majority of cases. The neural network AlphaFold that we developed was entered into the CASP14 assessment (May–July 2020; entered under the team name 'AlphaFold2' and a completely different model from our CASP13 AlphaFold system[10]). The CASP assessment is carried out biennially using recently solved structures that have not been deposited in the PDB or publicly disclosed so that it is a blind test for the participating methods, and has long served as the gold-standard assessment for the accuracy of structure prediction[25,26].

In CASP14, AlphaFold structures were vastly more accurate than competing methods. AlphaFold structures had a median backbone accuracy of 0.96 Å r.m.s.d.$_{95}$ (Cα root-mean-square deviation at 95% residue coverage) (95% confidence interval = 0.85–1.16 Å) whereas the next best performing method had a median backbone accuracy of 2.8 Å r.m.s.d.$_{95}$ (95% confidence interval = 2.7–4.0 Å) (measured on CASP domains; see Fig. 1a for backbone accuracy and Supplementary Fig. 14 for all-atom accuracy). As a comparison point for this accuracy, the width of a carbon atom is approximately 1.4 Å. In addition to very accurate domain structures (Fig. 1b), AlphaFold is able to produce highly accurate side chains when the backbone is highly accurate and considerably improves over template-based methods even when strong templates are available. The all-atom accuracy of AlphaFold was 1.5 Å r.m.s.d.$_{95}$ (95% confidence interval = 1.2–1.6 Å) compared with the 3.5 Å r.m.s.d.$_{95}$ (95% confidence interval = 3.1–4.2 Å) of the best alternative method. Finally, the model is able to provide precise, per-residue estimates of its reliability that should enable the confident use of these predictions.

The AlphaFold network

The AlphaFold network directly predicts the 3D coordinates of all heavy atoms for a given protein using the primary amino acid sequence and aligned sequences of homologues as inputs (see Methods for details of inputs including databases, MSA construction and use of templates).

The network comprises two main stages. First, the trunk of the network processes the inputs through repeated layers of a novel neural network block that we term Evoformer to produce an $N_{\text{seq}} \times N_{\text{res}}$ array (N_{seq}, number of sequences; N_{res}, number of residues) that represents a processed MSA and an $N_{\text{res}} \times N_{\text{res}}$ array that represents residue pairs. The key innovations in the Evoformer block are new mechanisms to exchange information within the MSA and pair representations that enable direct reasoning about the spatial and evolutionary relationships.

The trunk of the network is followed by the structure module that introduces an explicit 3D structure in the form of a rotation and translation for each residue of the protein (global rigid body frames). Both within the structure module and throughout the whole network, we reinforce the notion of iterative refinement by repeatedly applying the final loss to outputs and then feeding the outputs recursively into the same modules. The iterative refinement using the whole network (which we term 'recycling' and is related to approaches in computer vision[27,28]) contributes markedly to accuracy with minor extra training time (see Supplementary Methods 1.8 for details).

Evoformer

The key principle of the building block of the network—named Evoformer—is to view the prediction of protein structures as a graph inference problem in 3D space in which the edges of the graph are defined by residues in proximity. The elements of the pair representation encode information about the relation between the residues. The columns of the MSA representation encode the individual residues of the input sequence while the rows represent the sequences in which those residues appear.

Within the pair representation, there are two different update patterns. Both are inspired by the necessity of consistency of the pair representation—for a pairwise description of amino acids to be representable as a single 3D structure, many constraints must be satisfied including the triangle inequality on distances. On the basis of this intuition, we arrange the update operations on the pair representation in terms of triangles of edges involving three different nodes. In particular, we add an extra logit bias to axial attention[29] to include

the 'missing edge' of the triangle and we define a non-attention update operation 'triangle multiplicative update' that uses two edges to update the missing third edge (see Supplementary Methods 1.6.5 for details).

End-to-end structure prediction

The structure module operates on a concrete 3D backbone structure using the pair representation and the original sequence row (single representation) of the MSA representation from the trunk. The 3D backbone structure is represented as N_{res} independent rotations and translations, each with respect to the global frame (residue gas). These rotations and translations—representing the geometry of the N-Cα-C atoms—prioritize the orientation of the protein backbone so that the location of the side chain of each residue is highly constrained within that frame. Conversely, the peptide bond geometry is completely unconstrained and the network is observed to frequently violate the chain constraint during the application of the structure module as breaking this constraint enables the local refinement of all parts of the chain without solving complex loop closure problems. Satisfaction of the peptide bond geometry is encouraged during fine-tuning by a violation loss term. Exact enforcement of peptide bond geometry is only achieved in the post-prediction relaxation of the structure by gradient descent in the Amber[30] force field. Empirically, this final relaxation does not improve the accuracy of the model as measured by the global distance test (GDT)[31] or lDDT-Cα[32] but does remove distracting stereochemical violations without the loss of accuracy.

The residue gas representation is updated iteratively in two stages. First, a geometry-aware attention operation that we term 'invariant point attention' (IPA) is used to update an N_{res} set of neural activations (single representation) without changing the 3D positions, then an equivariant update operation is performed on the residue gas using the updated activations. The IPA augments each of the usual attention queries, keys and values with 3D points that are produced in the local frame of each residue such that the final value is invariant to global rotations and translations (see Methods 'IPA' for details).

Predictions of side-chain χ angles as well as the final, per-residue accuracy of the structure (pLDDT) are computed with small per-residue networks on the final activations at the end of the network. The estimate of the TM-score (pTM) is obtained from a pairwise error prediction that is computed as a linear projection from the final pair representation.

Discussion

AlphaFold has already demonstrated its utility to the experimental community, both for molecular replacement[33] and for interpreting cryogenic electron microscopy maps[34]. Moreover, because AlphaFold outputs protein coordinates directly, AlphaFold produces predictions in graphics processing unit (GPU) minutes to GPU hours depending on the length of the protein sequence (for example, around one GPU minute per model for 384 residues; see Methods for details). This opens up the exciting possibility of predicting structures at the proteome-scale and beyond—in a companion paper[35], we demonstrate the application of AlphaFold to the entire human proteome[35].

The explosion in available genomic sequencing techniques and data has revolutionized bioinformatics but the intrinsic challenge of experimental structure determination has prevented a similar expansion in our structural knowledge. By developing an accurate protein structure prediction algorithm, coupled with existing large and well-curated structure and sequence databases assembled by the experimental community, we hope to accelerate the advancement of structural bioinformatics that can keep pace with the genomics revolution. We hope that AlphaFold—and computational approaches that apply its techniques for other biophysical problems—will become essential tools of modern biology.

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Methods

IPA

The IPA module combines the pair representation, the single representation and the geometric representation to update the single representation (Supplementary Fig. 8). Each of these representations contributes affinities to the shared attention weights and then uses these weights to map its values to the output. The IPA operates in 3D space. Each residue produces query points, key points and value points in its local frame. These points are projected into the global frame using the backbone frame of the residue in which they interact with each other. The resulting points are then projected back into the local frame. The affinity computation in the 3D space uses squared distances and the coordinate transformations ensure the invariance of this module with respect to the global frame (see Supplementary Methods 1.8.2 'Invariant point attention (IPA)' for the algorithm, proof of invariance and a description of the full multi-head version).

Inputs and data sources

Inputs to the network are the primary sequence, sequences from evolutionarily related proteins in the form of a MSA created by standard tools including jackhmmer[36] and HHBlits[37], and 3D atom coordinates of a small number of homologous structures (templates) where available. For both the MSA and templates, the search processes are tuned for high recall; spurious matches will probably appear in the raw MSA but this matches the training condition of the network.

For MSA search on BFD + Uniclust30, and template search against PDB70, we used HHBlits[37] and HHSearch[38] from hhsuite v.3.0-beta.3 (version 14/07/2017). For MSA search on Uniref90 and clustered MGnify, we used jackhmmer from HMMER3[39]. For constrained relaxation of structures, we used OpenMM v.7.3.1[40] with the Amber99sb force field[30]. For neural network construction, running and other analyses, we used TensorFlow[41], Sonnet[42], NumPy[43], Python[44] and Colab[45].

Inference regimen

Using our CASP14 configuration for AlphaFold, the trunk of the network is run multiple times with different random choices for the MSA cluster centres (see Supplementary Methods 1.11.2 for details of the ensembling procedure). The full time to make a structure prediction varies considerably depending on the length of the protein. Representative timings for the neural network using a single model on V100 GPU are 4.8 min with 256 residues, 9.2 min with 384 residues and 18 h at 2,500 residues. These timings are measured using our open-source code, and the open-source code is notably faster than the version we ran in CASP14 as we now use the XLA compiler[46].

Since CASP14, we have found that the accuracy of the network without ensembling is very close or equal to the accuracy with ensembling and we turn off ensembling for most inference. Without ensembling, the network is 8× faster and the representative timings for a single model are 0.6 min with 256 residues, 1.1 min with 384 residues and 2.1 h with 2,500 residues.

Residues	With ensembling (CASP14)	Without ensembling
256	4.8 min	0.6 min
384	9.2 min	1.1 min
2,500	18 h	2.1 h

Table 1 | Inference times. One model on one V100 GPU, by protein length.

Compiled for this edition from the timings given in 'Inference regimen'.

Metrics

The predicted structure is compared to the true structure from the PDB in terms of lDDT metric[32], as this metric reports the domain accuracy without requiring a domain segmentation of chain structures. The distances are either computed between all heavy atoms (lDDT) or only the Cα atoms to measure the

backbone accuracy (lDDT-Cα). As lDDT-Cα only focuses on the Cα atoms, it does not include the penalty for structural violations and clashes. Domain accuracies in CASP are reported as GDT[31] and the TM-score[47] is used as a full chain global superposition metric.

We also report accuracies using the r.m.s.d.$_{95}$ (Cα r.m.s.d. at 95% coverage). We perform five iterations of (1) a least-squares alignment of the predicted structure and the PDB structure on the currently chosen Cα atoms (using all Cα atoms in the first iteration); (2) selecting the 95% of Cα atoms with the lowest alignment error. The r.m.s.d. of the atoms chosen for the final iterations is the r.m.s.d.$_{95}$. This metric is more robust to apparent errors that can originate from crystal structure artefacts, although in some cases the removed 5% of residues will contain genuine modelling errors.

Data availability

All input data are freely available from public sources. We show experimental structures from the PDB with accession numbers 6Y4F[48], 6YJ1[49], 6VR4[50], 6SK0[51], 6FES[52], 6W6W[53], 6T1Z[54] and 7JTL[55].

Code availability

Source code for the AlphaFold model, trained weights and inference script are available under an open-source license at https://github.com/deepmind/alphafold.

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